

19 In an investigation, a plant cell was placed in pure water.

The initial rate at which water molecules entered the cell, R , was greater than the initial rate at which water molecules left the cell.

In a second investigation, a plant cell of the same type was placed in a solution with a water potential equal to that of the cell contents.

What will happen in the second investigation over a period of five minutes?

- A** Water molecules will not enter or leave the cell because the water potential of the cell contents is equal to that of the solution.
- B** Water molecules will enter and leave the cell in equal amounts, both at an initial rate that is less than R in the first investigation.
- C** Water molecules will enter and leave the cell in equal amounts, both at an initial rate that is greater than R in the first investigation.
- D** Water molecules will enter and leave the cell in equal amounts, both at an initial rate that is equal to R in the first investigation.